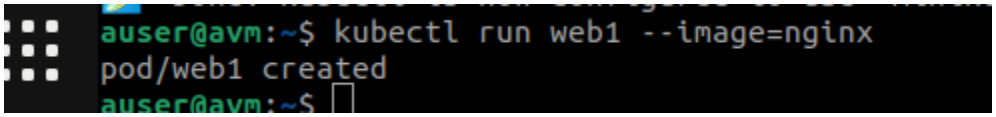
Assignment 6 (Cloud Services & Security)

PRN No. - 260840127015 Name - Khendake Pradip Goraksh Course - PGCP-HPCSA

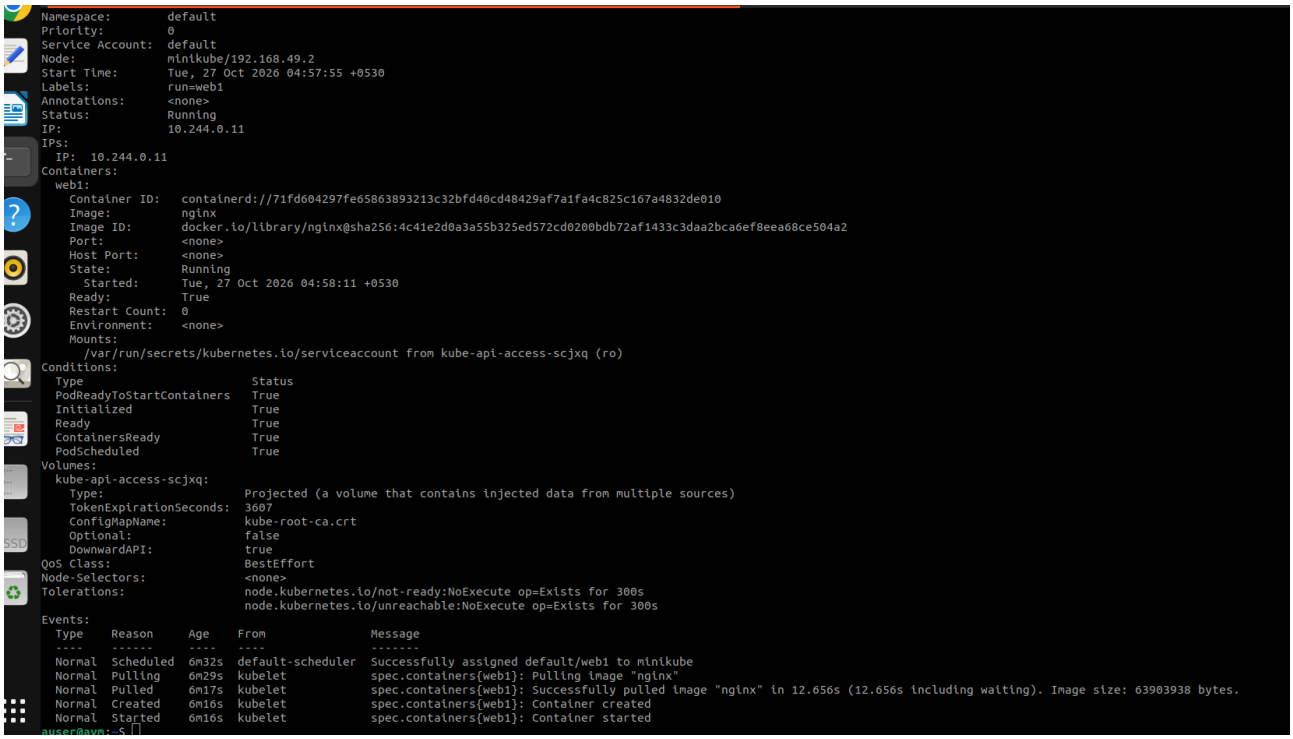
| 1 | Create a Pod named web1 using the nginx image.



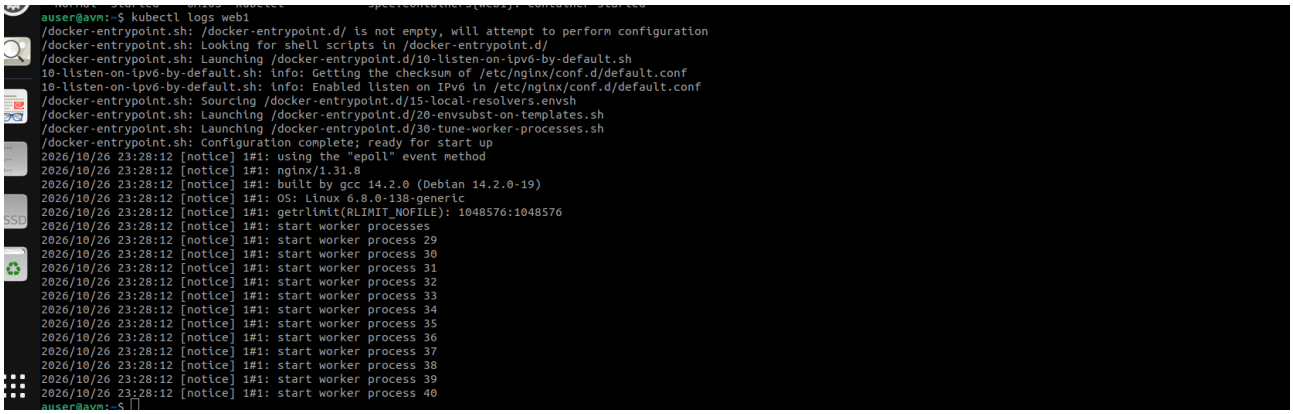
| 2 | Display all Pods running in the current namespace.



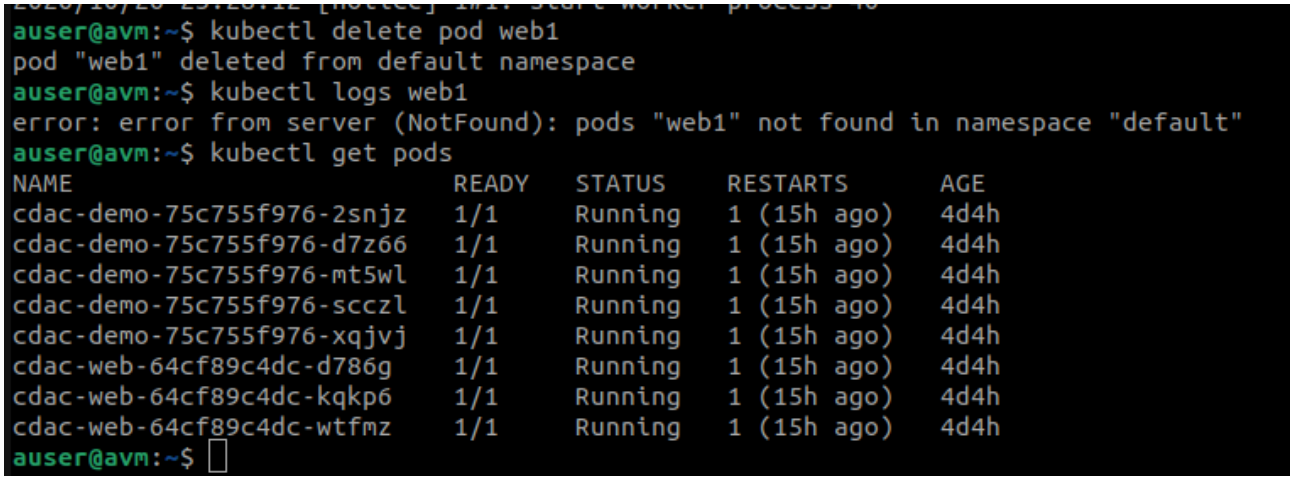
| 3 | Display detailed information about the Pod web1.



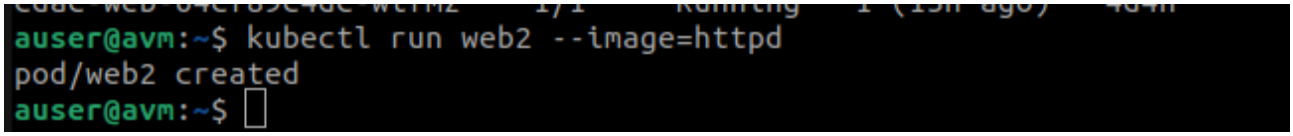
| 4 | Display the logs generated by the Pod web1.



| 5 | Delete the Pod web1 and verify that it has been deleted.



| 6 | Create a Pod named web2 using the httpd image.



| 7 | Create a ReplicaSet named web-rs with 3 replicas using the nginx image.

```
auser@avm:~$ cat <<EOF > web-rs.yaml
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: web-rs
spec:
  replicas: 3
  selector:
    matchLabels:
      app: web-rs
  template:
    metadata:
      labels:
        app: web-rs
    spec:
      containers:
      - name: nginx
        image: nginx
EOF
auser@avm:~$ ls -l web-rs.yaml
-rw-rw-r-- 1 auser auser 258 Oct 27 20:43 web-rs.yaml
auser@avm:~$ kubectl apply -f web-rs.yaml
replicaset.apps/web-rs created
auser@avm:~$
```

| 8 | Display the ReplicaSet and verify that 3 Pods have been created.

|

```
auser@avm:~$ kubectl get rs web-rs
NAME     DESIRED   CURRENT   READY   AGE
web-rs   3         3         3       13m
auser@avm:~$ kubectl get pods -l app=web-rs
NAME           READY   STATUS    RESTARTS   AGE
web-rs-fpwcn   1/1     Running   0          14m
web-rs-jtvk7   1/1     Running   0          14m
web-rs-tfb6d   1/1     Running   0          14m
auser@avm:~$
```

| 9 | Delete one Pod managed by web-rs and verify that Kubernetes creates a replacement Pod.

```
auser@avm:~$ kubectl get pods
NAME                          READY   STATUS    RESTARTS      AGE
cdac-demo-75c755f976-2snjz    1/1     Running   1 (16h ago)   4d4h
cdac-demo-75c755f976-d7z66    1/1     Running   1 (16h ago)   4d4h
cdac-demo-75c755f976-mt5wl    1/1     Running   1 (16h ago)   4d4h
cdac-demo-75c755f976-scczl    1/1     Running   1 (16h ago)   4d4h
cdac-demo-75c755f976-xqjvj    1/1     Running   1 (16h ago)   4d4h
cdac-web-64cf89c4dc-d786g     1/1     Running   1 (16h ago)   4d5h
cdac-web-64cf89c4dc-kqkp6     1/1     Running   1 (16h ago)   4d5h
cdac-web-64cf89c4dc-wtfmz     1/1     Running   1 (16h ago)   4d5h
web-rs-fpwcn                  1/1     Running   0             15m
web-rs-jtvk7                  1/1     Running   0             15m
web-rs-tfb6d                  1/1     Running   0             15m
web2                          1/1     Running   0             19m
auser@avm:~$ kubectl delete pod web-rs-jtvk7
pod "web-rs-jtvk7" deleted from default namespace
auser@avm:~$ kubectl get pods -l app=web-rs
NAME           READY   STATUS    RESTARTS   AGE
web-rs-894f9   1/1     Running   0          21s
web-rs-fpwcn   1/1     Running   0          16m
web-rs-tfb6d   1/1     Running   0          16m
auser@avm:~$
```

| 10 | Scale the ReplicaSet web-rs from 3 replicas to 5 replicas.

```
auser@avm:~$ kubectl get pods -l app=web-rs
NAME           READY   STATUS    RESTARTS   AGE
web-rs-894f9   1/1     Running   0          21s
web-rs-fpwcn   1/1     Running   0          16m
web-rs-tfb6d   1/1     Running   0          16m
auser@avm:~$ kubectl sacle rs web-rs --replicas=5
error: unknown command "sacle" for "kubectl"

Did you mean this?
        scale
auser@avm:~$ kubectl scale rs web-rs --replicas=5
replicaset.apps/web-rs scaled
auser@avm:~$ kubectl get pods -l app=web-rs
NAME           READY   STATUS              RESTARTS   AGE
web-rs-894f9   1/1     Running             0          94s
web-rs-fgszr   0/1     ContainerCreating   0          5s
web-rs-fpwcn   1/1     Running             0          17m
web-rs-tbq6s   0/1     ContainerCreating   0          5s
web-rs-tfb6d   1/1     Running             0          17m
auser@avm:~$
```

| 11 | Reduce the number of replicas in web-rs from 5 to 2.

```
auser@avm:~$ kubectl get pods -l app=web-rs
NAME           READY   STATUS              RESTARTS   AGE
web-rs-894f9   1/1     Running             0          94s
web-rs-fgszr   0/1     ContainerCreating   0          5s
web-rs-fpwcn   1/1     Running             0          17m
web-rs-tbq6s   0/1     ContainerCreating   0          5s
web-rs-tfb6d   1/1     Running             0          17m
auser@avm:~$ kubectl scale rs web-rs --replicas=2
replicaset.apps/web-rs scaled
auser@avm:~$ kubectl get pods -l app=web-rs
NAME           READY   STATUS    RESTARTS   AGE
web-rs-fpwcn   1/1     Running   0          18m
web-rs-tfb6d   1/1     Running   0          18m
auser@avm:~$
```

| 12 | Create a Deployment named web-deploy using the nginx image with 3 replicas.

```
auser@avm:~$ kubectl create deployment web-deploy --image=nginx --replicas=3
deployment.apps/web-deploy created
auser@avm:~$ kubectl get pods -l app=web-deploy
NAME                          READY   STATUS              RESTARTS   AGE
web-deploy-b8488874-c8x7f     0/1     ContainerCreating   0          7s
web-deploy-b8488874-jr756     0/1     ContainerCreating   0          7s
web-deploy-b8488874-qbktm     1/1     Running             0          7s
auser@avm:~$
```

| 13 | Display the Deployment, ReplicaSet and Pods associated with web-deploy.

```
auser@avm:~$ kubectl get deployment,rs,pods -l app=web-deploy
NAME                             READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/web-deploy       3/3     3            3           6m53s

NAME                                        DESIRED   CURRENT   READY   AGE
replicaset.apps/web-deploy-b8488874         3         3         3       6m53s

NAME                                READY   STATUS    RESTARTS   AGE
pod/web-deploy-b8488874-c8x7f       1/1     Running   0          6m53s
pod/web-deploy-b8488874-jr756       1/1     Running   0          6m53s
pod/web-deploy-b8488874-qbktm       1/1     Running   0          6m53s
auser@avm:~$
```

| 14 | Scale the Deployment web-deploy from 3 replicas to 5 replicas.

```
auser@avm:~$ kubectl scale deployment web-deploy --replicas=5
deployment.apps/web-deploy scaled
auser@avm:~$ kubectl get pods -l app=web-deploy
NAME                          READY   STATUS    RESTARTS   AGE
web-deploy-b8488874-4ts6t     1/1     Running   0          29s
web-deploy-b8488874-c8x7f     1/1     Running   0          8m48s
web-deploy-b8488874-jr756     1/1     Running   0          8m48s
web-deploy-b8488874-qbktm     1/1     Running   0          8m48s
web-deploy-b8488874-rzgv9     1/1     Running   0          29s
auser@avm:~$
```

| 15 | Update the image used by web-deploy from nginx to httpd and verify the rollout.

```
auser@avm:~$ kubectl set image deployment/web-deploy *=httpd
deployment.apps/web-deploy image updated
auser@avm:~$ kubectl rollout status deployment/web-deploy
deployment "web-deploy" successfully rolled out
auser@avm:~$
```

| 16 | Deliberately create a crashing Pod using the busybox image with the command sh -c "exit 1". Observe its status and restart count.

```
auser@avm:~$ ###############16################
auser@avm:~$ kubectl run crash-pod --image=busybox --restart=Always -- sh -c "exit 1"
pod/crash-pod created
auser@avm:~$ kubectl get pods crash-pod
NAME        READY   STATUS   RESTARTS   AGE
crash-pod   0/1     Error    0          23s
auser@avm:~$
```

| 17 | Create a Pod using the busybox image that runs sh -c "sleep 5; exit 1". Observe how its status changes over time.

```
auser@avm:~$ kubectl run sleep-pod --image=busybox --restart=Always -- sh -c "sleep 5; exit 1"
pod/sleep-pod created
auser@avm:~$ kubectl get pods sleep-pod -w
NAME        READY   STATUS             RESTARTS      AGE
sleep-pod   0/1     Error              0             12s
sleep-pod   1/1     Running            1 (3s ago)    13s
sleep-pod   0/1     Error              1 (9s ago)    19s
sleep-pod   0/1     CrashLoopBackOff   1 (12s ago)   30s
sleep-pod   1/1     Running            2 (15s ago)   33s
sleep-pod   0/1     Error              2 (20s ago)   38s
sleep-pod   0/1     CrashLoopBackOff   2 (25s ago)   62s
sleep-pod   1/1     Running            3 (28s ago)   65s
sleep-pod   0/1     Error              3 (33s ago)   70s
sleep-pod   0/1     CrashLoopBackOff   3 (52s ago)   2m2s
sleep-pod   1/1     Running            4 (54s ago)   2m4s
sleep-pod   0/1     Error              4 (61s ago)   2m11s
```

| 18 | Create a Pod using the nginx image but specify a deliberately incorrect image name, such as nginx-invalid. Observe the Pod status and events.

```
auser@avm:~$ ##########################17############################
auser@avm:~$ kubectl run bad-image-pod --image=nginx-invalid
pod/bad-image-pod created
auser@avm:~$ kubectl describe pod bad-image-pod
Name:             bad-image-pod
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Tue, 27 Oct 2026 21:54:34 +0530
Labels:           run=bad-image-pod
Annotations:      <none>
Status:           Pending
IP:               10.244.0.31
IPs:
  IP:  10.244.0.31
Containers:
  bad-image-pod:
    Container ID:
    Image:          nginx-invalid
    Image ID:
    Port:           <none>
    Host Port:      <none>
    State:          Waiting
      Reason:       ErrImagePull
    Ready:          False
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-8bq22 (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       False
  ContainersReady             False
  PodScheduled                True
Volumes:
  kube-api-access-8bq22:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
```

| 19 | Delete a Pod belonging to web-deploy and verify that the Deployment creates a replacement Pod.

```
auser@avm:~$ ##################19#################
auser@avm:~$ kubectl get pods -l app=web-deploy
NAME                          READY   STATUS    RESTARTS   AGE
web-deploy-7fbd466958-9p26x   1/1     Running   0          42m
web-deploy-7fbd466958-hplr5   1/1     Running   0          42m
web-deploy-7fbd466958-mrn7d   1/1     Running   0          42m
web-deploy-7fbd466958-p5jcs   1/1     Running   0          42m
web-deploy-7fbd466958-rp7gl   1/1     Running   0          42m
auser@avm:~$ kubectl delete pod web-deploy-7fbd466958-hplr5
pod "web-deploy-7fbd466958-hplr5" deleted from default namespace
auser@avm:~$ kubctl get pods -l app=web-deploy -w
Command 'kubctl' not found, did you mean:
  command 'kubectl' from snap kubectl (1.35.9)
See 'snap info <snapname>' for additional versions.
auser@avm:~$ kubectl get pods -l app=web-deploy -w
NAME                          READY   STATUS    RESTARTS   AGE
web-deploy-7fbd466958-9p26x   1/1     Running   0          43m
web-deploy-7fbd466958-l7jwj   1/1     Running   0          55s
web-deploy-7fbd466958-mrn7d   1/1     Running   0          43m
web-deploy-7fbd466958-p5jcs   1/1     Running   0          43m
web-deploy-7fbd466958-rp7gl   1/1     Running   0          43m
```

| 20 | Delete the Deployment web-deploy and verify that its associated ReplicaSet and Pods are also deleted.

```
^C
auser@avm:~$ kubectl delete deployment web-deploy
deployment.apps "web-deploy" deleted from default namespace
auser@avm:~$ kubectl get deployments,rs,pods -l app=web-deploy
No resources found in default namespace.
auser@avm:~$
```